

CNC Machine Operation SOP

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1. Purpose

This SOP establishes the safe operating procedure for Computer Numerical Control (CNC) machines including CNC lathes, milling centers, and machining centers. It is designed to prevent injuries from moving parts, flying chips and coolant, tool breakage, and unexpected machine motion, and to ensure consistent, quality-compliant machining operations.

2. Scope

Applies to all certified CNC operators, setters, and maintenance technicians working on CNC equipment within the machine shop. This procedure covers pre-operation checks, operation, tool changes, and shutdown.

3. Pre-Operation Checks

- Verify that the machine guard interlocks, emergency stop buttons, and light curtains are functional before starting any job.
- Inspect coolant levels, hydraulic and pneumatic pressure gauges, and lubrication system for correct readings.
- Confirm that the correct program has been loaded and verify the tool offsets and work coordinate system against the job traveler.
- Perform a dry run (air cut) of a new or modified program with the door closed and feed override reduced, to check for collision paths before actual cutting.
- Ensure the workpiece is securely clamped using the correct fixture, and that clamps do not obstruct the tool path.

4. Operating Procedure

The operator must remain at the control panel or within immediate reach of the emergency stop at all times while the machine is cutting. Machine doors and guards must remain closed during the entire cutting cycle; interlocks must never be bypassed or defeated under any circumstance, including for troubleshooting. Only trained and certified operators are authorized to override feed rates or spindle speeds, and only within limits specified in the job's process sheet. Loose clothing, jewelry, and unrestrained long hair are prohibited near the machine due to entanglement risk with rotating spindles and chucks.

5. Tool Changing and Setup

- Machine must be brought to a complete stop with the spindle at zero RPM before any manual tool change.
- Use the designated tool-change key or automatic tool changer (ATC) procedure only; never force a tool into the holder.
- Verify tool length and diameter offsets are re-entered into the control after any tool change or tool break.

- Remove all setup tools, wrenches, and measuring instruments from the machine bed before resuming the cycle — a documented tool-accountability check is mandatory.

6. Housekeeping and Chip Management

Chips and swarf must be cleared using a brush, rake, or the machine's built-in conveyor — never by hand, and never with compressed air directed at the operator or bystanders. Coolant spills must be cleaned immediately to prevent slip hazards. Machine surroundings must be kept clear of obstructions to allow immediate access to the emergency stop and safe evacuation routes.

7. Emergency Procedures

In the event of abnormal noise, vibration, smoke, or a tool breakage, the operator must immediately press the emergency stop and notify the shift supervisor and maintenance team. The machine must not be restarted until a qualified maintenance technician has inspected and cleared it. Any near-miss, however minor, must be reported through the incident reporting system on the same shift.

8. Maintenance and Lockout

Before any maintenance, cleaning, or adjustment activity that requires access beyond the guarded zone, the machine must be fully isolated following the plant's Lockout-Tagout (LOTO) procedure (see the Lockout-Tagout SOP). Only authorized maintenance personnel may remove machine guards, and only after verifying zero energy state.

9. Training and Certification

Only operators who have completed the CNC certification program, including a documented practical assessment, are authorized to operate these machines unsupervised. Recertification is required every 2 years or after any CNC-related incident.